

WEEK WISE AGENDA (weeks 1 through 3,4)

Week 1: Embeddings & Retrieval Foundations

- Study contextual embedding models for a huge medical corpus (survey a few — e.g. sentence-transformers, OpenAI/Gemini embedding endpoints) and compare representational quality, testing different retrieval strategies.
- Implement text retrieval (like BM25) as a baseline.
- Implement semantic retrieval using vector similarity — cosine similarity search over embeddings (via MongoDB or any other software).
- Combine both (hybrid retrieval: sparse + dense) and evaluate retrieval quality.
- Implement and evaluate re-ranking of retrieved chunks to improve precision on top-k results.

Week 2: Graph RAG for Medical Use Cases

- Study Graph RAG architecture and where it improves over vector-only RAG.
- Automate the graph construction pipeline.
- Identify and document deployment complexity / bottlenecks for Graph RAG in a clinical setting.

Week 3: Agentic RAG, Query Understanding & Integration

- Learn agentic RAG patterns — multi-step retrieval, tool-calling, self-correction/routing between retrieval strategies based on query type.
- Implement query translation / query rewriting: handle abbreviations, shorthand, and underspecified clinical queries by expanding them with context before retrieval (e.g. LLM-based query rewriting step, or a medical abbreviation-expansion layer).